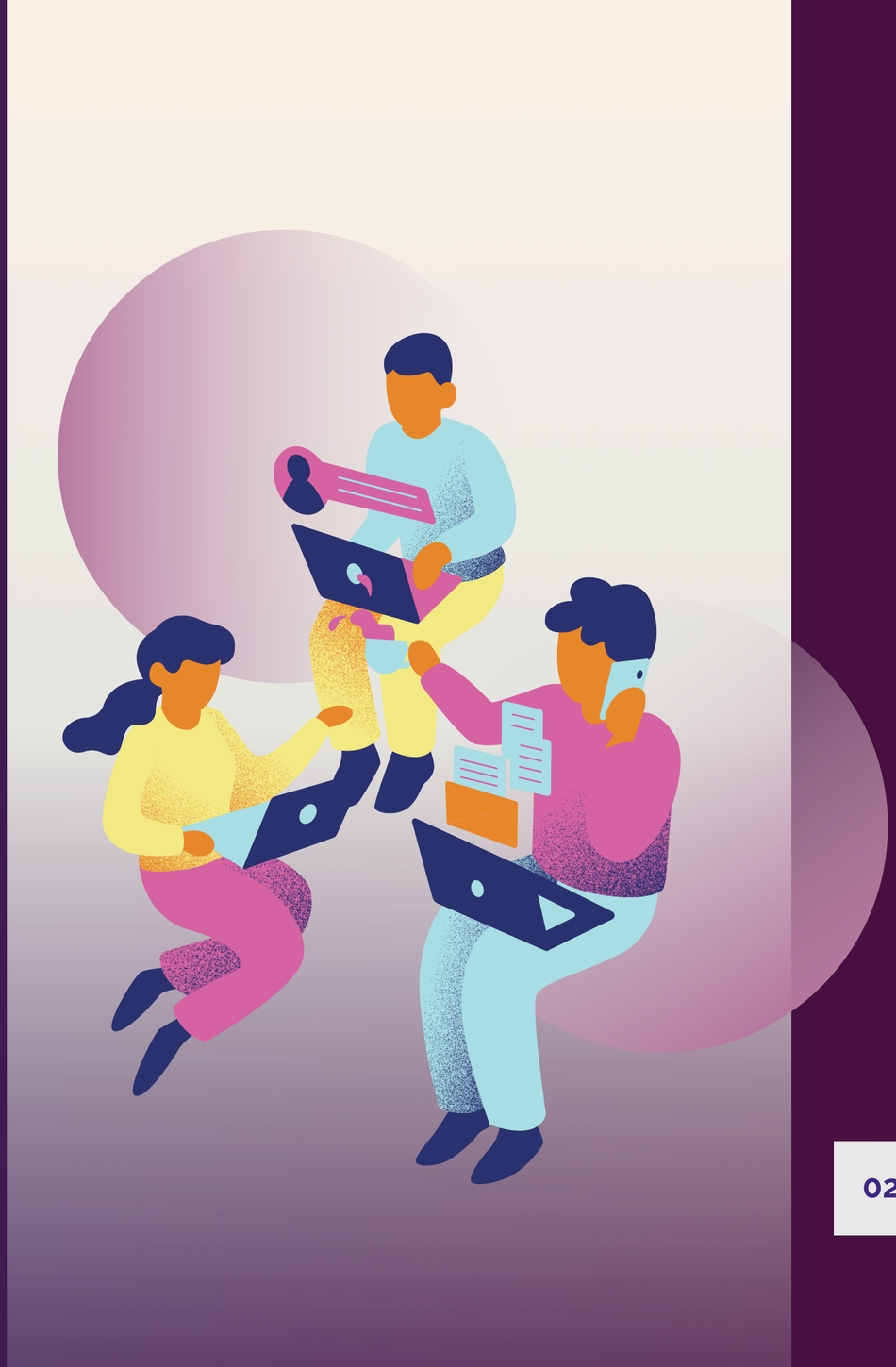
DESIGN AND IMPLEMENTATION OF AN ENTERPRISE DATA INTELLIGENCE PLATFORM FOR UNIFIED ANALYTICS



PROJECT AREA: DATABASE SYSTEMS AND APPLICATIONS
FUNCTIONAL DOMAIN: CONSUMER GOODS AND RETAIL ANALYTICS

PROBLEM STATEMENT

Retail organizations generate data across sales, inventory, procurement, supply chain, customer management and finance.
Data is distributed across heterogeneous systems and formats.
Fragmented data leads to:
Inconsistent reporting
Limited cross-functional visibility
Delayed decision-making
Traditional BI often requires SQL knowledge and predefined reporting workflows.



PROPOSED SOLUTION

- Build a unified analytical environment for enterprise retail data.
- Enable business-user self-service analytics through semantic AI.
- Improve query accuracy, usability and decision-making efficiency.

PROPOSED ARCHITECTURE

BUSINESS INTELLIGENCE

Visible data analysis and dashboards



INTELLIGENCE & ANALYTICS

Advanced AI and semantic analysis

DATA INTEGRATION & STORAGE

Centralized cloud data lake

DATA WAREHOUSE

Structured storage for analytical data

DATA SOURCES

Diverse origins of raw data

DATA SOURCE DESCRIPTIONS

Data Source Strategy

The platform uses a synthetic retail dataset designed to represent the operational data generated by a supermarket or consumer retail enterprise. The dataset supports the development and validation of data ingestion, incremental processing, centralized storage, and AI-assisted analytics capabilities without exposing confidential business information.

Source Types

Structured File Sources: CSV datasets containing customer, product, inventory, promotion, purchase order, sales, store, and vendor information.

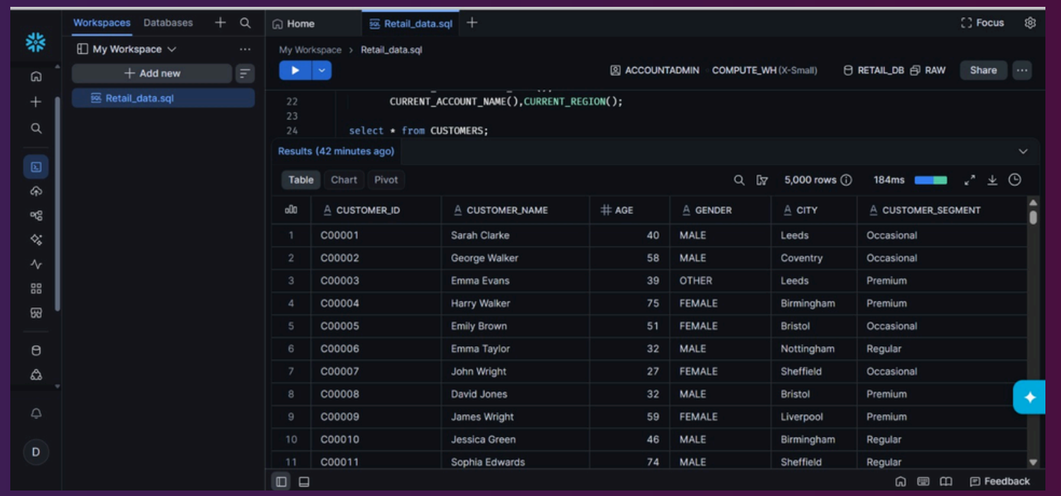
REST API Source: An API-based retail sales data source used to demonstrate programmatic data retrieval, API integration, and ingestion of incremental records into the data platform.

Incremental Data Files: Separate customer and product update datasets used to simulate changes in existing records and the arrival of new records over time.

Multiple Data Formats: The architecture is designed to accommodate structured file formats such as CSV, Excel, and Parquet, alongside API-based ingestion.

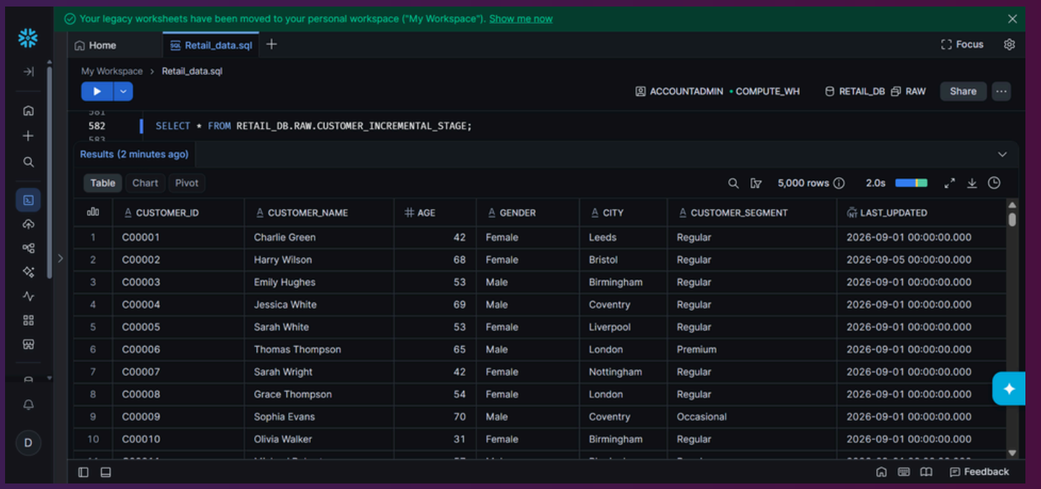
CSV RAW DATA LOADED FOR CUSTOMER

customers	23-09-2026 21:57	Microsoft Excel Com...	232 KB
inventory	23-09-2026 21:57	Microsoft Excel Com...	1,929 KB
products	23-09-2026 21:57	Microsoft Excel Com...	66 KB
promotions	23-09-2026 21:57	Microsoft Excel Com...	6 KB
purchase_orders	23-09-2026 21:57	Microsoft Excel Com...	2,308 KB
README	23-09-2026 21:57	Text Document	2 KB
sales	23-09-2026 21:57	Microsoft Excel Com...	2,631 KB
stores	23-09-2026 21:57	Microsoft Excel Com...	3 KB
vendors	23-09-2026 21:57	Microsoft Excel Com...	11 KB



CSV INCREMENTAL TRUNCATE AND LOAD FOR CUSTOMER BASED ON LAST UPDATED DATE

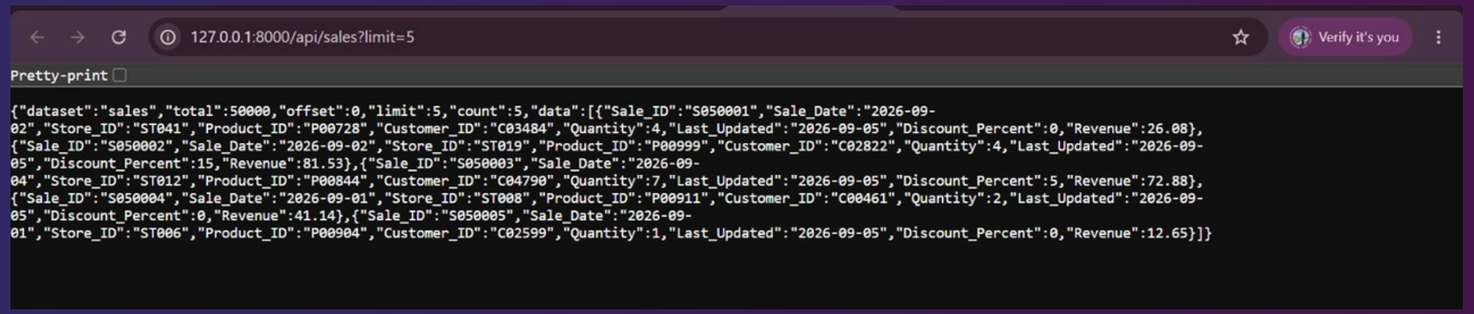
customers_incremental	07-10-2026 09:16	Microsoft Excel Com...	285 KB
inventory_incremental	07-10-2026 09:16	Microsoft Excel Com...	3,004 KB
products_incremental	07-10-2026 09:16	Microsoft Excel Com...	78 KB
promotions_incremental	07-10-2026 09:16	Microsoft Excel Com...	3 KB
purchase_orders_incremental	07-10-2026 09:16	Microsoft Excel Com...	2,844 KB
README_INCREMENTAL_LOAD	07-10-2026 09:16	Text Document	2 KB
sales_incremental	07-10-2026 09:16	Microsoft Excel Com...	2,943 KB
stores_incremental	07-10-2026 09:16	Microsoft Excel Com...	3 KB
vendors_incremental	07-10-2026 09:16	Microsoft Excel Com...	13 KB

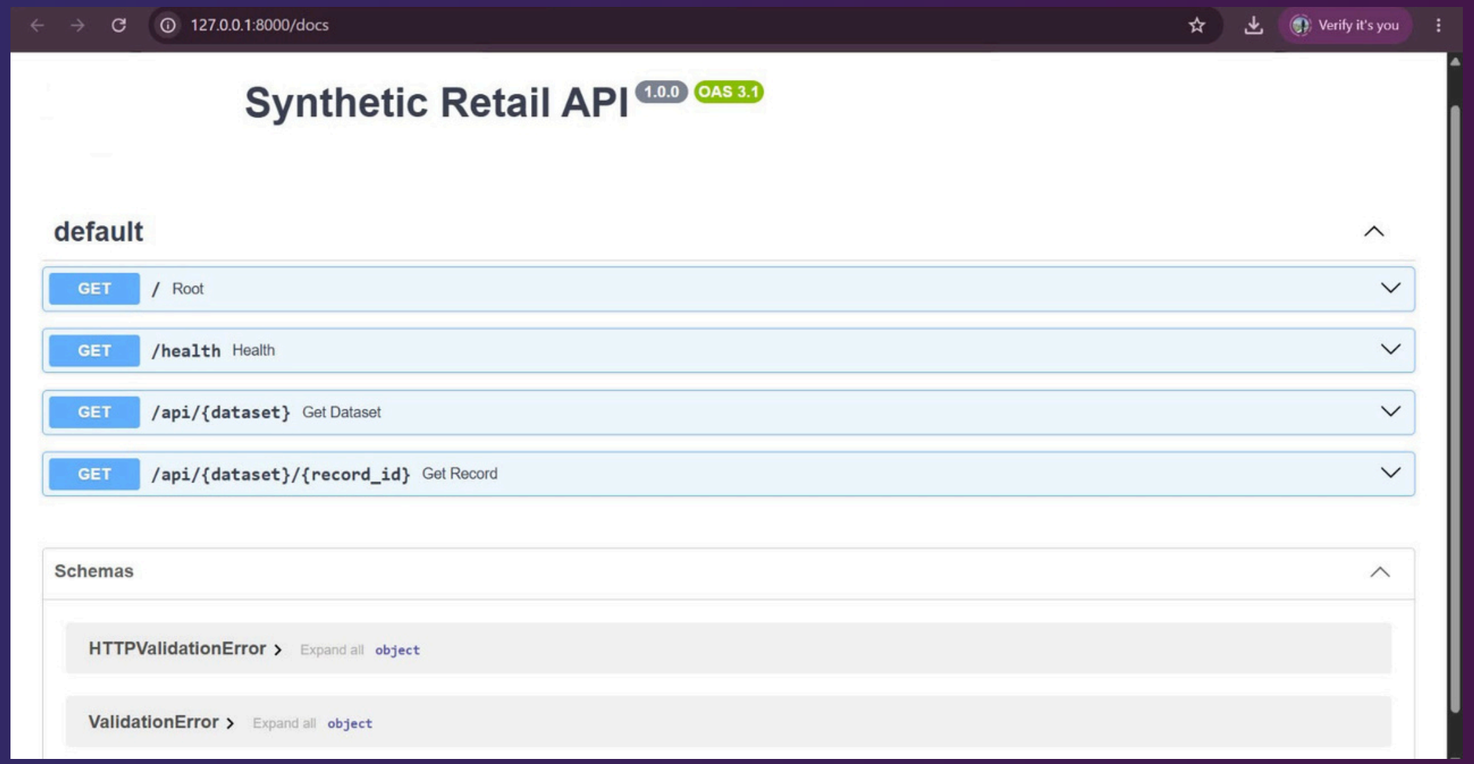


API DATA SOURCE

REST API created and tested

Set up a FastAPI REST API for synthetic retail sales data.





DATASET AND DATA INGESTION

RETAIL ENTITIES

The platform currently includes eight logical retail entities across customer, product, and six additional retail datasets.

TABLE_NAME	TOTAL_ROWS	UNIQUE_IDS
INVENTORY	50000	50000
PROMOTIONS	150	150
PURCHASE_ORDERS	50000	50000
SALES	50000	50000
STORES	50	50
VENDORS	200	200
CUSTOMERS	5000	5000
PRODUCTS	1000	1000

AZURE DATA FACTORY IMPLEMENTATION

Metadata-driven bulk loading

A reusable pipeline named PL_Retail_Bulk_Load was configured to process multiple retail entities through a shared workflow.
Processing steps

Benefits

- Reduces repetitive pipeline configuration.
- Supports multiple entities through a common workflow.
- Simplifies the addition of new entities.
- Improves pipeline maintainability through metadata-driven orchestration.

Processing steps

- A Lookup activity reads the retail configuration containing folder paths, filenames, and destination table names.
- A ForEach activity iterates through the configuration records.
- Dynamic expressions pass the current source folder and filename to the Copy activity.
- The destination table name is passed dynamically to the Snowflake V2 dataset.
- Data is loaded into the corresponding Snowflake raw tables.

INCREMENTAL DATA PROCESSING

Customer and product workflows

Separate processing workflows have been developed for customers and products.

Key capabilities

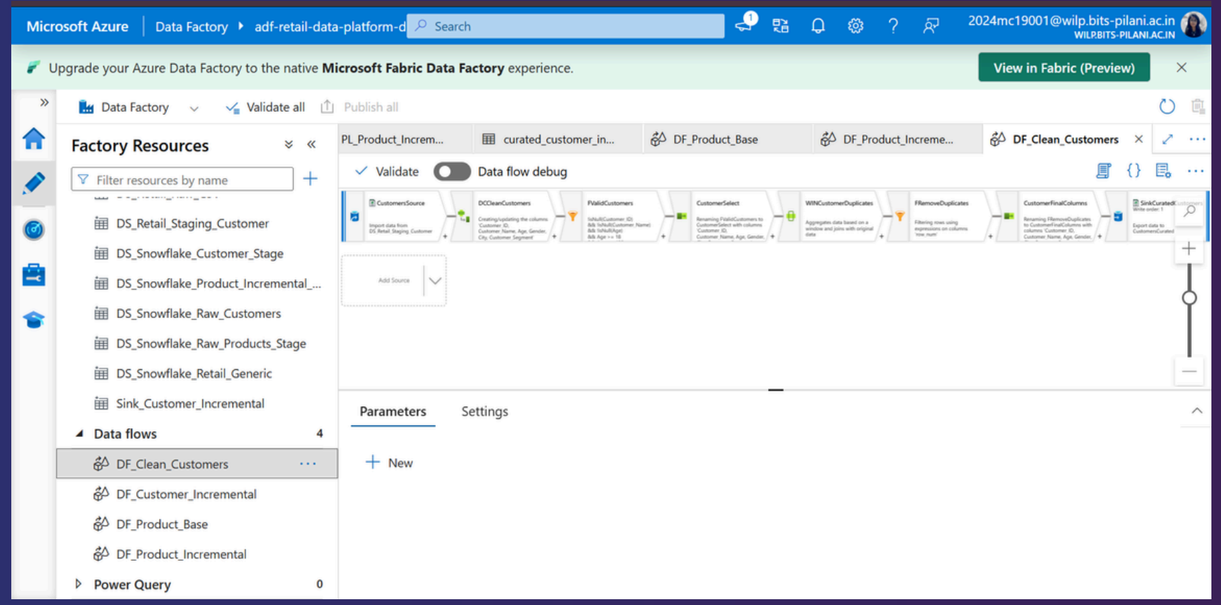
- Separates initial loading from incremental processing.
- Supports the incorporation of new and updated records.
- Uses database-side merge logic to maintain final tables.
- Reduces dependence on repeatedly loading the entire dataset.

Processing sequence

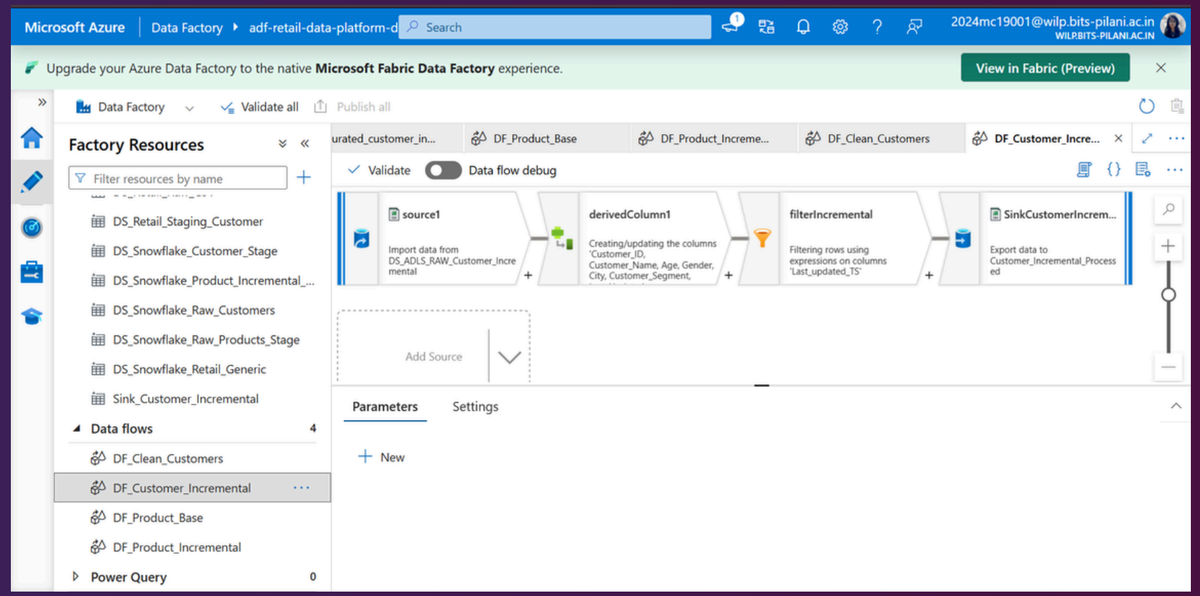
- Load the initial dataset into the raw layer.
- Receive incremental source data containing new or updated records.
- Apply filtering and transformation logic.
- Load the processed incremental data into the appropriate destination.
- Execute a Snowflake stored procedure to merge the changes into the final table.
- Automate execution through configured ADF triggers.

The incremental workflows are currently implemented for customer and product data. The other six retail entities have been loaded into the raw layer and remain candidates for subsequent incremental-processing development.

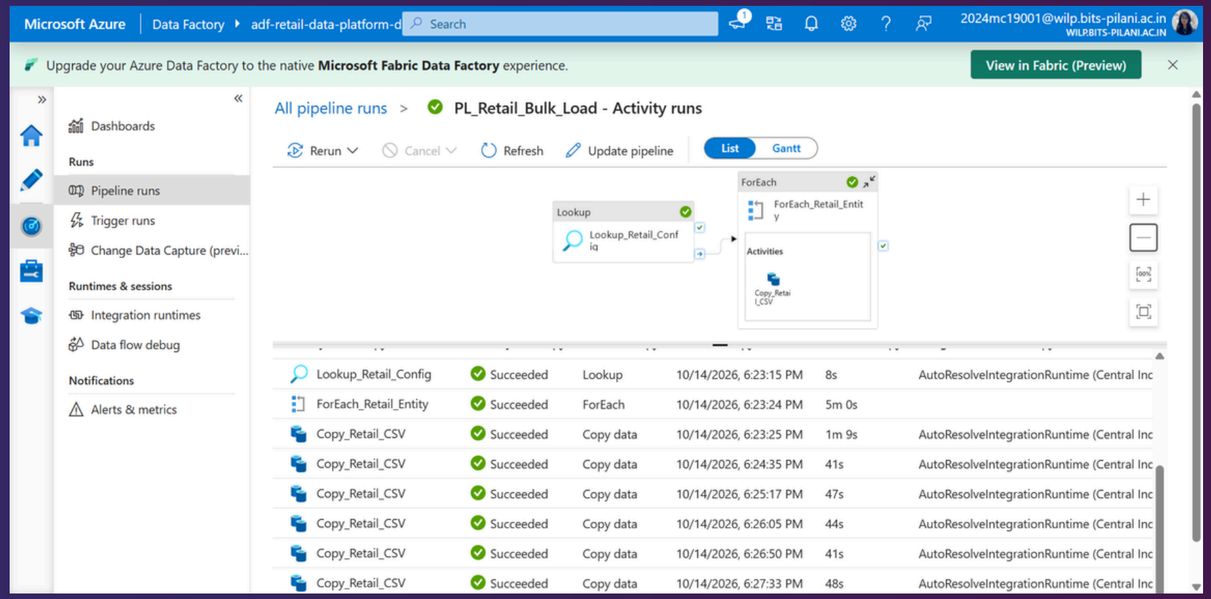
Clean data for customer Dataflow



Incremental data load for customer Dataflow



ForEach load, to load all raw files at once



SNOWFLAKE DATA MANAGEMENT AND VALIDATION

Centralized data storage

Snowflake is used to store retail datasets and execute data-processing and analytical queries.
The current implementation includes:

- Raw tables for customer and product data.
- Final customer and product tables maintained through stored procedures.
- Six additional raw retail tables.
- SQL-based validation of table structures and record counts.

Initial validation results

- Six additional retail tables were successfully populated.
- The six tables contain 150,400 records in total.
- Primary identifier uniqueness checks passed for the six tables.
- Table column names and data types were inspected through Snowflake metadata queries.

Next validation activities

- Check null values across relevant columns.
- Validate foreign-key relationships between retail entities.
- Verify incremental processing and rerun behavior.
- Evaluate query performance and data-processing efficiency.

AI-POWERED ANALYTICS USING SNOWFLAKE CORTEX

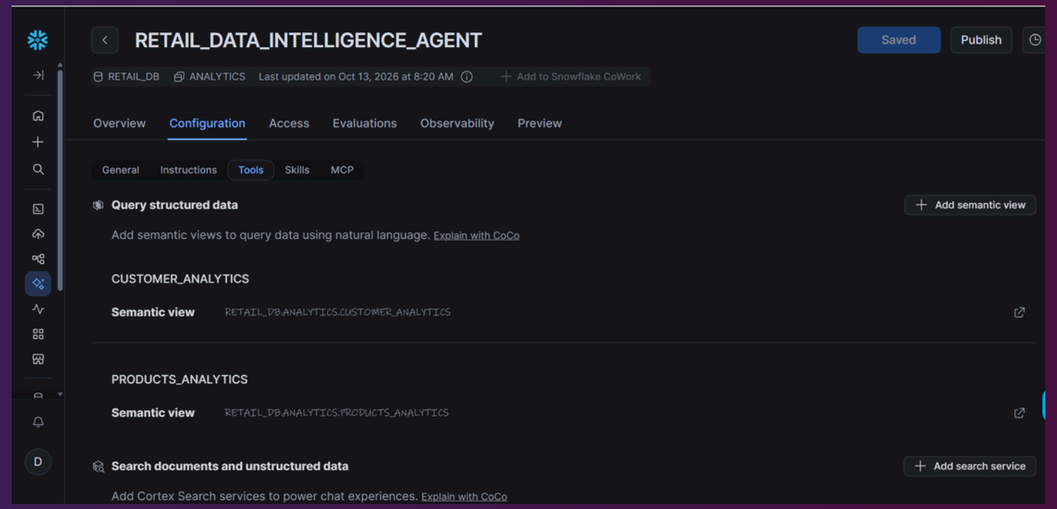
AI AGENT IMPLEMENTATION

A Snowflake Cortex AI agent has been configured for customer and product analytics using separate semantic analysts.



Customer semantic analyst

- Provides business context for customer-related questions, including:
- Customer counts and demographic analysis.
- Customer segment distributions.
- Geographic customer analysis.
- Age-related analytical questions.



Product semantic analyst

- Provides business context for product-related questions, including:
- Product counts and category analysis.
- Brand-level comparisons.
- Average product prices and costs.
- Individual product analysis.

CURRENT PROGRESS AND FUTURE WORK

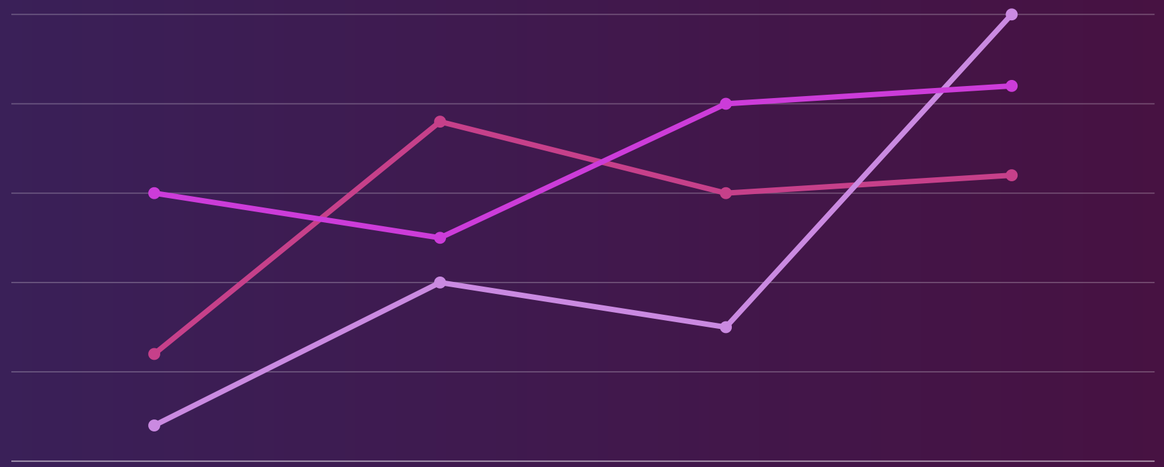
Completed

- Initial customer and product raw data ingestion.
- Customer and product incremental processing workflows.
- Snowflake stored procedures for merging incremental changes.
- Configured automated pipeline triggers.
- Metadata-driven loading of six additional retail entities.
- Verification of the six tables' record counts and primary identifier uniqueness.
- Snowflake Cortex AI agent with customer and product semantic analysts.

Planned work

- Complete data-quality and relationship validation across retail entities.
- Extend transformation and incremental-processing capabilities where required.
- Strengthen the semantic analysts with additional verified questions and business rules.
- Evaluate AI answer accuracy, usability, and decision efficiency against traditional BI.
- Implement the Power BI executive dashboard.
- Document performance, security, limitations, and final evaluation findings.

Project Progress — Completed, In Progress, and Pending Tasks



THANK YOU!

DATA ANALYSIS IS KEY TO BUSINESS GROWTH AND SUCCESS!